

Rudin, *Principles of Mathematical Analysis*

Chapter 7 — Sequences and Series of Functions

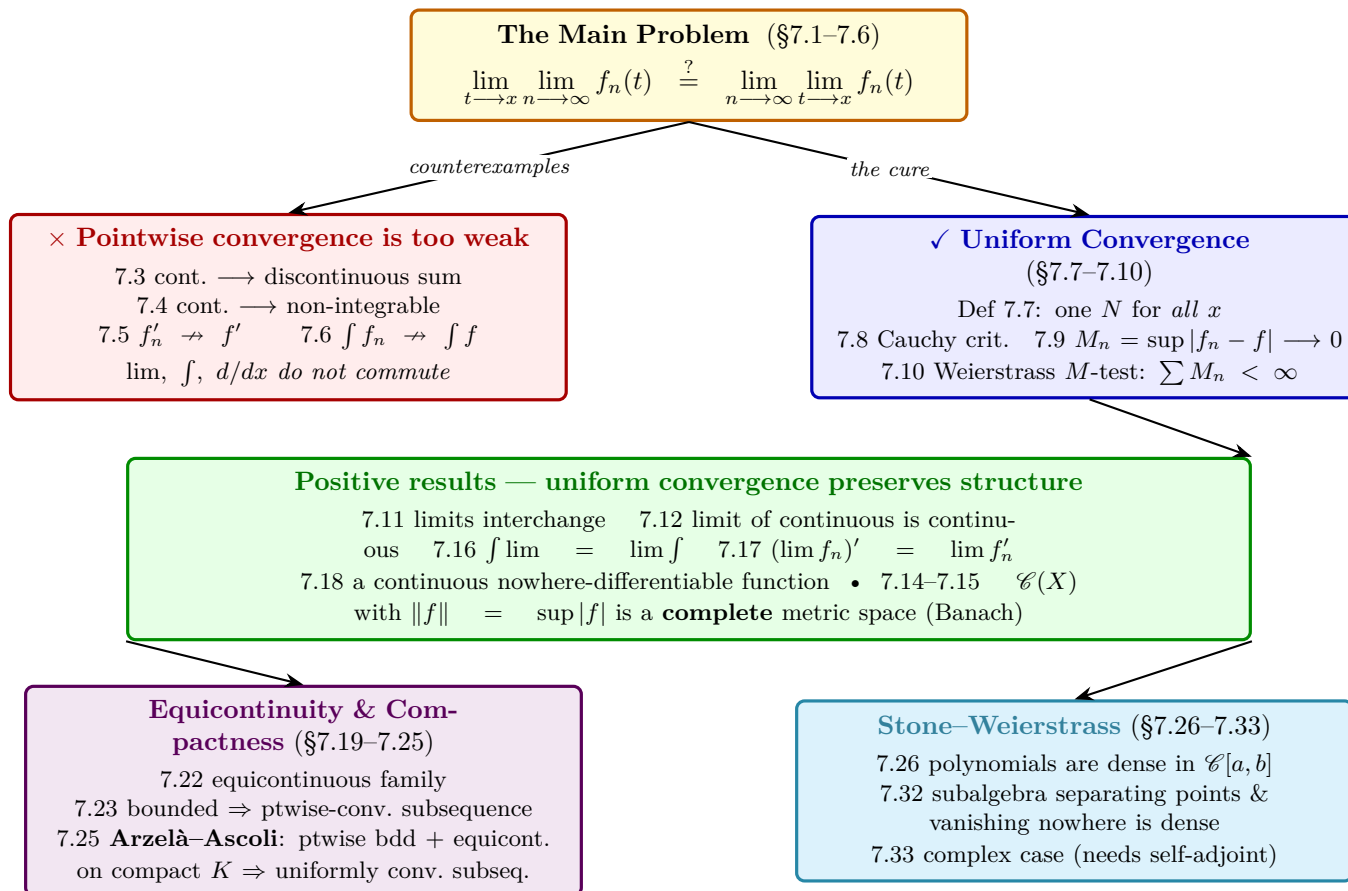
A High-Level Overview

Abstract

The whole chapter answers *one* question: when we take a limit of functions ($f = \lim f_n$ or $f = \sum f_n$), are the good properties (continuity, integrability, differentiability) preserved? Equivalently, may two limit processes be interchanged? Pointwise convergence is *not* enough (four counterexamples); the cure is **uniform convergence**, after which everything is repaired. The chapter culminates in the Arzelà–Ascoli and Stone–Weierstrass theorems.

Visual Overview — the logical skeleton of Chapter 7

Core question: when may two limit processes be interchanged?



Book pp. 143–171 • builds toward Ch. 8 (power series, e^x , trig, Fourier, Γ).

1 The Main Problem (§7.1–7.6)

Definition 1.1 (Pointwise convergence). A sequence $\{f_n\}$ defined on a set E *converges pointwise* to f if

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) \quad (x \in E),$$

and the *sum* of a series is $f(x) = \sum_{n=1}^{\infty} f_n(x)$.

The central question is whether limit operations commute. Asking whether a limit of continuous functions is continuous is exactly asking whether

$$\lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t) \stackrel{?}{=} \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t). \quad (1)$$

Pointwise convergence is too weak — four counterexamples

7.2 (double sequence)

With $s_{m,n} = \frac{m}{m+n}$,

$$\lim_{n \rightarrow \infty} \lim_{m \rightarrow \infty} s_{m,n} = 1 \neq 0 = \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} s_{m,n}.$$

7.3 (continuous $\rightarrow$ discontinuous)

For $f(x) = \sum_{n=0}^{\infty} \frac{x^2}{(1+x^2)^n}$ a convergent series of continuous functions has the discontinuous sum

$$f(x) = \begin{cases} 0, & x = 0, \\ 1 + x^2, & x \neq 0. \end{cases}$$

7.4 (continuous $\rightarrow$ non-integrable)

$f(x) = \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} (\cos m! \pi x)^{2n} = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}, \end{cases}$ the Dirichlet function — not Riemann integrable.

7.5 (derivative fails)

$f_n(x) = \frac{\sin nx}{\sqrt{n}} \rightarrow 0 = f$, but $f'_n(x) = \sqrt{n} \cos nx$, so $f'_n(0) = \sqrt{n} \rightarrow \infty \neq f'(0) = 0$.

7.6 (integral fails)

$f_n(x) = n^2 x(1-x^2)^n$ on $[0, 1]$ has $f_n \rightarrow 0$, yet

$$\int_0^1 f_n(x) dx = \frac{n^2}{2n+2} \rightarrow \infty \neq \int_0^1 \lim_n f_n(x) dx = 0.$$

Conclusion: $\lim$, $\int$, and $\frac{d}{dx}$ do *not* commute in general.

2 Uniform Convergence (§7.7–7.10)

Definition 2.1 (Uniform convergence). $\{f_n\} \rightarrow f$ *uniformly* on E if for every $\varepsilon > 0$ there is an integer N (depending on ε *only*, the *same* for all x) such that

$$n \geq N \implies |f_n(x) - f(x)| \leq \varepsilon \quad \text{for all } x \in E.$$

The whole point: in pointwise convergence N may depend on x ; in uniform convergence one N works simultaneously for every x .

Theorem 2.2 (Cauchy criterion, 7.8). $\{f_n\}$ *converges uniformly* on E iff for every $\varepsilon > 0$ there is N such that

$$m \geq N, n \geq N, x \in E \implies |f_n(x) - f_m(x)| \leq \varepsilon.$$

Theorem 2.3 (Sup test, 7.9). If $M_n = \sup_{x \in E} |f_n(x) - f(x)|$, then $f_n \rightarrow f$ *uniformly* on E iff $M_n \rightarrow 0$ as $n \rightarrow \infty$.

Theorem 2.4 (Weierstrass M -test, 7.10). If $|f_n(x)| \leq M_n$ for all $x \in E$ and $\sum M_n < \infty$, then $\sum f_n$ *converges uniformly* on E .

3 Positive Results and the Space $\mathcal{C}(X)$ (§7.11–7.18)

Uniform convergence repairs every counterexample of §1:

Theorem 3.1 (Interchange of limits, 7.11). If $f_n \rightarrow f$ *uniformly* on E , x a limit point of E , and $\lim_{t \rightarrow x} f_n(t) = A_n$, then $\{A_n\}$ *converges* and

$$\lim_{t \rightarrow x} f(t) = \lim_{n \rightarrow \infty} A_n, \quad \text{i.e.} \quad \lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t).$$

Theorem 3.2 (7.12, 7.16, 7.17).

- (7.12) A uniform limit of continuous functions is continuous.
- (7.16) If $f_n \rightarrow f$ *uniformly* on $[a, b]$ and $f_n \in \mathcal{R}(\alpha)$, then $\int_a^b f \, d\alpha = \lim_{n \rightarrow \infty} \int_a^b f_n \, d\alpha$.
- (7.17) If f_n are differentiable, $\{f_n(x_0)\}$ *converges* for some x_0 , and $\{f'_n\}$ *converges uniformly*, then $f_n \rightarrow f$ *uniformly* and $f'(x) = \lim_{n \rightarrow \infty} f'_n(x)$.

Theorem 3.3 (A continuous nowhere-differentiable function, 7.18). There exists a real continuous function on $\mathbb{R}$ which is nowhere differentiable, e.g. $f(x) = \sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n \varphi(4^n x)$, where φ is the distance from x to the nearest integer (convergence is uniform by the M -test).

Definition 3.4 (The metric space $\mathcal{C}(X)$, 7.14). For a metric space X , let $\mathcal{C}(X)$ be the set of all complex-valued, continuous, *bounded* functions on X , with the *supremum norm*

$$\|f\| = \sup_{x \in X} |f(x)|, \quad \|f + g\| \leq \|f\| + \|g\|.$$

The distance $\|f - g\|$ makes $\mathcal{C}(X)$ a metric space, and convergence in this metric is *exactly* uniform convergence.

Theorem 3.5 (Completeness, 7.15). $\mathcal{C}(X)$ with the sup-norm metric is a **complete** metric space (a Banach space).

4 Equicontinuity and Arzelà–Ascoli (§7.19–7.25)

Definition 4.1 (Equicontinuity, 7.22). A family $\mathcal{F}$ of functions on E is *equicontinuous* if for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon \quad \text{for every } f \in \mathcal{F}.$$

(One δ works for the whole family at once.)

Theorem 4.2 (7.23). If $\{f_n\}$ is pointwise bounded on a countable set E , then it has a subsequence $\{f_{n_k}\}$ converging at every point of E .

Theorem 4.3 (Arzelà–Ascoli, 7.25). If K is compact, $f_n \in \mathcal{C}(K)$, and $\{f_n\}$ is pointwise bounded and equicontinuous, then

- (a) $\{f_n\}$ is uniformly bounded on K ;
- (b) $\{f_n\}$ contains a uniformly convergent subsequence.

This is the function-space analogue of Bolzano–Weierstrass.

5 The Stone–Weierstrass Theorem (§7.26–7.33)

Theorem 5.1 (Weierstrass approximation, 7.26). If f is continuous on $[a, b]$, there is a sequence of polynomials P_n with $P_n \rightarrow f$ uniformly on $[a, b]$. Equivalently, polynomials are dense in $\mathcal{C}[a, b]$.

Definition 5.2 (Algebra, separates points, vanishes nowhere; 7.28–7.30). A family $\mathcal{A}$ is an *algebra* if it is closed under $f + g$, fg , and cf . It *separates points* if for $x_1 \neq x_2$ some $f \in \mathcal{A}$ has $f(x_1) \neq f(x_2)$; it *vanishes at no point* if for each x some $f \in \mathcal{A}$ has $f(x) \neq 0$.

Theorem 5.3 (Stone–Weierstrass, real case, 7.32). Let $\mathcal{A}$ be an algebra of real continuous functions on a compact set K . If $\mathcal{A}$ separates points of K and vanishes at no point of K , then the uniform closure of $\mathcal{A}$ is all of $\mathcal{C}(K, \mathbb{R})$.

Theorem 5.4 (Complex case, 7.33). For a complex algebra the same conclusion holds provided $\mathcal{A}$ is self-adjoint (closed under complex conjugation). Without this hypothesis the theorem fails.

Where this leads

Chapter 7 is the methodological hub of the book. Its tools — uniform convergence, the M -test, completeness of $\mathcal{C}(X)$, and Stone–Weierstrass — are used in **Chapter 8** to define e^x , $\sin$, $\cos$, $\log$, the Γ function via power series, and to develop Fourier series.